Part	Topic
1	Vector, norms, dot product, cross product
2	Matrials: addition, multiplication, transcript
3	Line: plane, hyperplane and dimensionally
4	Eigenvalue, eigenvectors, convariance matrix
5	summary of result

1. Vectors :-

vectors: an orderds list of numbers with both magnitute and direction. A student's scores from a vector in R^5

$S = (math, Physic, chemistry, english, comsci)$

Norm - 1 :- This sum of absoulte values of the components

$$||x||_1 = |x^1| + |x|^2 + \ldots + |x_m|$$

Norm - 2 :- The straight - line lenght of the vector from the origin

$$||x_2|| = sqr(x_1^2 + x_2^2 + \ldots + x_m^2)$$

Angle between vectors :- This angle whose cosine is the product dilived by the product of norms small angle mean similar score profile

$$theta = arccos(a.b) / (||a||)(||b||)$$

Page
Date

cross product : Defined only R^3. It givens a vectors perpendicular to both inputs : its lenght equals the areea of the parallelogram they span.

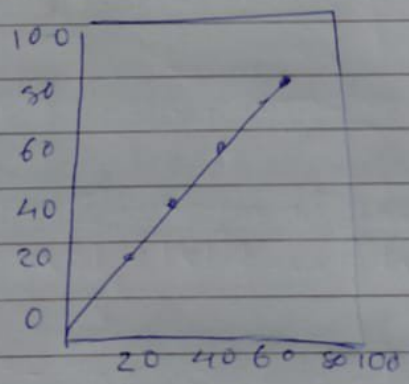
$a \times b = (a^3b^2 - a^3b^1 - a^1b^3, a^1 \cdot ab^2 - a^2b^1)$

in our dataset : for cmat, pys, che) $s_1 \times s_2 = \ell$

vector projection : The compent of a along the direction of b

$proj_3(a) = (a.b)/(a.b))b$

in our dataset . projection coefficient of s_1 on s_2 is the leftover residual is perpendicular to s_2



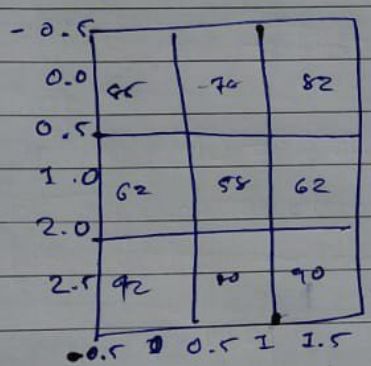
2. Matrics :-

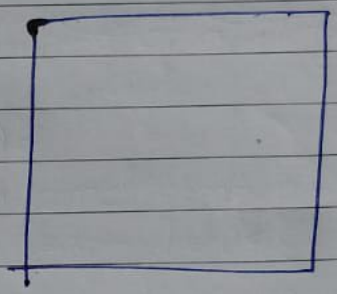
matrix : A rectangular array of numbers is with m row and n colums. our data matrix A is 30 x 5

matrix additions : Elements - wise additions : both matrix must have the same shapes.

Transpose :- Swaps rows and columns
$(A^T)_{ij} = A_{ji}$ $(AB)^T = B^T A^T$

inverse :- for a square matrix with non-zero determinant the matrix M^{-1} satisfies $M M^{-1} = I$ NON square matrix have no inverse $A^\dagger = (A^T A)^{-1} A^T$





3. Line, Plane, Hyperplane and Dimensionality

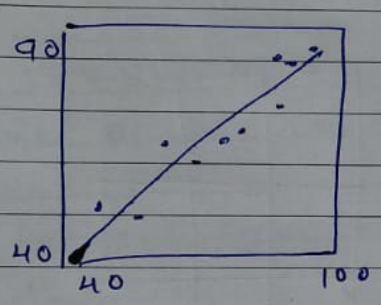
Line : A one-dimensional flat object in R^2 defined one equation

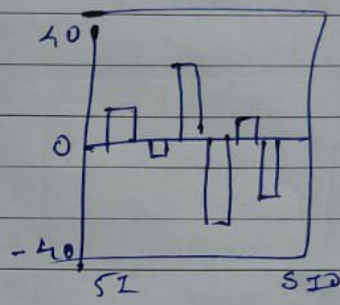
$y = mx + c$

plane :- A two-dimensional flat surface in R^3 defined by one linear equation

$ax + by + cz = d$

Dimensionally growth :- in R^3 a hyperplane is a line (1-D) in R^3 a plane (2-D) : in R^n a 3-D hyperplane : in R^n an (n-1) - dimensional one. Each additional subject adds one axis to the space one dimension to its hyperplane.





4. Eigenvalues, Eigenvector and pca

Convariance matrix: A symmetic matrix where entry (i, j) the covariance between variable 1 and variable j. The diagonals holds variances.

properties used: for symmtric eigenvalues are real. eigenvectors are otherwised sum of eigenvalues = trace(c), product = det(c)

principal componet Analysis: projects data onto the eigenvector matrix. The eigenvalue equals the variance captured along the eigenvector

5. Summery of Results :-

Concepts	Results
Norm 1 / Norm - 2	405 / 181.75
$S_1 . S_2$ and angle	26124.10.95 diy
$S_1 \times S_2$ (M, P, &)	(314, -441, 94)
det of 3×3 sub-matrix	-1460
Line fit	pysic = 0.1917 maths + 4.186
Largest eigvenvalue	11680.2 (93.67-)
PC 1 meaning	overall sci / tee ability
PC 2 meaning	English vs sci (5.01-)

End